CHAPTER 1

INTRODUCTION

Background of the Study

Many bakeries today still rely on manual methods in managing sales, inventory, and product supplies, which can lead to delays, inaccurate records, and product shortages. As customer demand continues to increase, bakery businesses need a more efficient and organized system to monitor daily transactions and supply chain operations. This study focuses on developing a web-based POS and supply chain management system that can help bakeries improve operational efficiency, reduce errors, and provide better service to customers. In today's fast-changing business environment, companies are continuously seeking innovative ways to improve operational efficiency, reduce costs, and enhance customer satisfaction. One of the most important aspects of business operations is the effective management of inventory, sales, and product distribution. Businesses, especially small and medium enterprises (SMEs), often face challenges in monitoring stocks, tracking sales transactions, and managing the movement of products from suppliers to customers. Traditional methods of inventory recording and manual transaction processing are time-consuming, prone to human error, and inefficient in handling large amounts of data. Because of these issues, many businesses are now adopting computerized systems that can automate processes and provide accurate, real-time information for decision-making. A Point of Sale (POS) system is a computerized solution that allows businesses to process sales transactions, monitor inventory, and generate reports efficiently. POS systems have become widely used in retail stores, supermarkets, restaurants, bakeries, and other commercial establishments because they simplify daily operations and improve transaction accuracy.

However, while many existing POS systems focus mainly on sales processing and inventory monitoring, some businesses still experience difficulties in managing their overall supply chain processes. Supply chain management involves the coordination of suppliers, inventory, warehousing, distribution, and customer demand to ensure that products are available at the right time and quantity. Without an effective supply chain management system, businesses may suffer from stock shortages, overstocking, delayed deliveries, and inaccurate inventory records. In the Philippines, many local businesses, particularly small retail stores and bakeries, still rely on manual methods or outdated systems in handling their inventory and supply chain operations. These traditional practices often result in inconsistencies in stock records, delays in order fulfillment, and losses caused by expired or missing products. Furthermore, manual recording consumes a significant amount of time and effort, limiting the productivity of employees and affecting the overall performance of the business. As customer demand continues to increase and competition becomes more intense, businesses need modern technological solutions that can streamline operations and improve efficiency. The integration of POS technology with supply chain management functions can significantly improve business processes. A POS-based Supply Chain Management System allows business owners and employees to monitor sales transactions, inventory levels, supplier information, and product movement in real-time. Through automation, businesses can reduce manual work, minimize human errors, and improve the accuracy of inventory tracking. In addition, the system can generate reports and analytics that help managers make informed decisions regarding restocking, sales trends, and product demand. By having accurate and updated information, businesses can avoid unnecessary expenses and improve customer satisfaction through faster and more reliable service.

Moreover, the development of Batch Flow is highly relevant in today's digital era where businesses are transitioning from manual operations to automated systems. Technological advancements have made it possible for businesses to adopt affordable and efficient information systems that improve operational performance. By implementing a POS-based supply chain management system, businesses can enhance productivity, reduce operational costs, and improve the accuracy of business records. The system can also help business owners monitor product availability, prevent inventory shortages, and maintain proper stock levels to meet customer demands effectively. Furthermore, the study may benefit business owners, employees, customers, and future researchers. Business owners may benefit from improved decision-making and better inventory control, while employees may experience easier and faster transaction processing. Customers may also benefit from improved service quality and product availability. Future researchers can use this study as a reference for developing more advanced business management systems and improving existing technologies related to POS and supply chain management. In conclusion, the increasing demand for efficient business operations and accurate inventory management emphasizes the need for an integrated POS-based supply chain management solution. The proposed Batch Flow system aims to provide businesses with a reliable and efficient platform for managing sales, inventory, and supply chain processes. By automating and centralizing business operations, the system can contribute to improved productivity, reduced errors, and enhanced customer satisfaction. Therefore, this study seeks to design and develop a POS-Based Supply Chain Management System that can support businesses in achieving more efficient and organized operations in the modern business environment.

Problem Statement

There are a lot of factors here, lack of proper communication between subordinates and higher ups can result in some system proposal getting canceled due to not accepting the system. Others are can be because people who work there are not into the technology. Bakeries are getting less of a recognition because of newer more popular bakeshops like Goldilocks or the “oven on wheels” style of marketing made bakeshops essentially getting less recognition. The study aims to develop Batch Flow: A POS-Based Supply Chain Management System for a bakery business to improve sales transactions, inventory management, and supply chain operations.

Manual and Time-Consuming Sales Transactions

Many bakery businesses still rely on manual methods such as handwritten receipts and calculators when processing customer transactions. This process consumes a significant amount of time, especially during busy business hours when there are many customers waiting in line. Manual computation also increases the possibility of human errors in pricing, quantity, and change calculation. Inaccurate transactions may lead to customer dissatisfaction and financial losses for the business. Because of these issues, the bakery needs a POS-based system that can automate transactions and improve the speed and accuracy of sales processing.

Inaccurate Inventory Monitoring

The bakery faces challenges in monitoring the availability of raw materials and finished products due to the use of manual inventory recording. Employees may forget to update stock records, resulting in discrepancies between the actual inventory and recorded data. This problem can lead to overstocking, product wastage, and shortages of essential ingredients needed for production. Inaccurate inventory tracking also affects the bakery’s ability to meet customer demand consistently. Therefore, an automated inventory management feature is necessary to provide real-time stock monitoring and maintain accurate inventory records.

Lack of Real-Time Supply Chain Tracking

The current bakery system does not provide real-time monitoring of supplier transactions, deliveries, and purchase orders. Because of this, management experiences difficulty in tracking the movement of supplies from suppliers to the bakery. Delays in deliveries and unrecorded supply transactions may disrupt production schedules and product availability. Without a centralized system, communication between suppliers and the bakery becomes less efficient and harder to manage. The proposed system aims to improve supply chain operations by providing organized and real-time tracking of supply-related activities.

Difficulty in Generating Sales and Inventory Reports

The bakery currently prepares reports manually, which requires employees to collect and organize records from different sources. This process is time-consuming and may result in incomplete or inaccurate reports. Delayed report generation affects the ability of management to make immediate and informed business decisions. Inaccurate reports may also impact financial monitoring, inventory planning, and sales analysis. By implementing the proposed system, the bakery can automatically generate accurate sales and inventory reports in a faster and more efficient manner.

Limited Data Organization and Security

Manual record-keeping makes it difficult for the bakery to organize and secure important business information. Paper-based records are vulnerable to loss, damage, and unauthorized access, which may affect the reliability of business data. Inconsistent data storage can also cause duplication of records and difficulty in retrieving important information when needed. Without proper security measures, sensitive sales and inventory information may be exposed to unauthorized individuals. The proposed POS-based supply chain management system will help improve data organization, storage, and security through a centralized and computerized database system.

Objectives

The main purpose of Batch Flow: A POS-Based Supply Chain Management System is to help bakeries manage their sales, inventory, suppliers, and transactions efficiently through an integrated web-based platform. The system aims to automate daily bakery operations to reduce manual work, minimize errors, and improve the accuracy of sales and inventory records. It also provides real-time monitoring of product stocks, orders, and deliveries to ensure smooth supply chain operations. Additionally, the system supports faster decision-making by generating organized reports and transaction summaries for managers and administrators. Overall, the system is designed to improve operational efficiency, customer service, and business productivity in bakery management.

1. To automate the bakery's sales and inventory management processes using a web-based POS system.

2. To monitor and track product stocks, supplies, and deliveries in real time.

3. To improve transaction accuracy and reduce manual recording errors.

4. To provide organized reports for sales, inventory, and supply chain activities.

5. To help managers and administrators make better business decisions through accurate and updated information.

Scope

Batch Flow: A POS-Based Supply Chain Management System is a web-based system designed to improve the daily operations of a bakery business by integrating point-of-sale and supply chain management processes into one platform. The system focuses on managing sales transactions, product inventory, supplier information, and purchase orders. It allows cashiers to process customer purchases efficiently while automatically updating inventory records in real time. Through this system, bakery staff can monitor product availability, reduce stock shortages, and improve transaction accuracy.

The system also includes administrative and managerial functions that support business monitoring and decision-making. Administrators can manage master data such as products, categories, prices, and user accounts, while managers can access sales reports, inventory summaries, and supplier transaction records. The system generates organized reports that help track bakery performance, sales trends, and supply activities. Additionally, the supplier module helps record deliveries and purchase transactions to maintain proper coordination between the bakery and suppliers.

Furthermore, the system is accessible through a web browser, making it convenient for authorized users within the bakery to access operational data. It is intended for small to medium-sized bakery businesses that need a more organized and automated way of handling sales and supply chain operations. The study mainly covers the development, implementation, and testing of the web-based system, including its usability and functionality in supporting bakery operations.

Limitations

- Limited only to bakery-related operations.
- Requires an internet connection.
- No online payment integration.

Significance of the Study

The proposed system, Batch Flow: A POS-Based Supply Chain Management System, is significant because it helps improve the efficiency, accuracy, and overall management of bakery operations. The system integrates Point-of-Sale (POS) functions with supply chain management processes, allowing bakery businesses to monitor sales, inventory, supplies, and transactions in real time. Through automation, the system reduces manual work, minimizes errors, and improves the speed of daily operations. It also provides organized records and reports that can support better decision-making for the business.

Bakery Owners and Management will benefit from the system because it helps them monitor sales, inventory levels, and product performance efficiently. The system provides accurate reports and real-time updates that support better decision-making and business planning. It also helps reduce losses caused by overstocking, shortages, and human errors in manual recording. By improving operational efficiency, the system can contribute to higher productivity and profitability for the bakery business.

Employees and Cashiers will benefit from the system through faster and more organized transaction processing. The POS feature simplifies sales recording, order management, and inventory updates, reducing workload and minimizing mistakes during operations. The system also helps staff perform their tasks more efficiently because information is centralized and easy to access. This can improve employee productivity and service quality inside the bakery.

Customers will benefit from the system because it improves the speed and accuracy of service. Transactions become faster, reducing waiting time during purchases. The system also helps ensure product availability since inventory is monitored properly. As a result, customers can experience better service satisfaction and a more convenient buying experience.

Future Researchers and Students may use this study as a reference for developing similar systems related to POS and supply chain management. The study can provide ideas, methodologies, and insights that may help improve future research and technological innovations in business management systems. It can also contribute to academic learning in the fields of Information Technology and Business Management.

Target Users

Admin: The admin is responsible for managing and controlling the overall system. The admin can create and manage user accounts, monitor system activities, update product information, and generate reports. They also ensure that the system operates properly and securely. Through the system, the admin can oversee all bakery operations in one centralized platform.

Cashier: The cashier uses the Point-of-Sale (POS) feature to process customer purchases and transactions. They are responsible for recording sales, issuing receipts, and handling payments accurately and quickly. The system helps cashiers reduce manual errors and speed up transaction processing. It also automatically updates inventory after every sale.

Manager: The manager monitors the performance and operations of the bakery. They use the system to review sales reports, inventory status, product demand, and employee performance. The manager can analyze business data to support decision-making and improve operational efficiency. The system also helps managers track supplies and monitor business growth.

Supplier: The suppliers are responsible for providing raw materials and bakery supplies needed by the business. Through the system, suppliers can receive supply requests, monitor orders, and coordinate deliveries with the bakery. The system helps improve communication and ensures timely restocking of ingredients and materials.

Customer: The customers are the buyers of bakery products and services. They benefit from faster and more accurate transactions through the POS system. Customers can experience improved service quality, reduced waiting time, and better product availability because inventory is properly monitored. The system helps provide a more convenient and satisfying purchasing experience.

CHAPTER 2

RELATED LITERATURE AND RELATED SYSTEMS

This chapter on Related Literature presents different studies, systems, and technologies related to Batch Flow: A POS-Based Supply Chain Management System. It discusses concepts and existing literature about Point-of-Sale systems, inventory management, and supply chain management used in bakery businesses and similar industries. The purpose of this chapter is to provide supporting information and references that help strengthen the development of the proposed system. It also highlights the importance of automated business processes in improving efficiency, accuracy, and overall operational performance.

Related Literature

Duarte et. al (2021), effective production and inventory policies are essential in bakery businesses to improve resource allocation and product quality. Furthermore, the study emphasized that optimized inventory management helps reduce waste and operational costs in multiproduct bakery operations. This literature is relevant to Batch Flow because the proposed system integrates POS and supply chain management for efficient inventory monitoring and production planning.

As stated by Xu, Huang, Weng, and Do (2021), integrated food supply chains in the baking industry improve customer service and operational efficiency through online ordering systems. Moreover, the researchers stated that digital platforms help bakeries respond to changing customer demands during online transactions. This study supports Batch Flow because the proposed system also uses web technologies to improve bakery operations and customer transactions.

Bolarinwa and Fajebe (2021), explained that inventory optimization improves bakery performance by balancing service levels, delivery lead time, and inventory turnover. Consequently, the implementation of inventory management systems helps bakeries reduce shortages and unnecessary stock accumulation. This literature is connected to Batch Flow because the proposed system focuses on accurate stock monitoring and inventory control.

Monteiro, Moita Neto, and Silva (2021), emphasized that sustainable management practices in bakeries improve operational efficiency and resource utilization. Additionally, the study highlighted that technology-based management systems support better monitoring of bakery operations and environmental sustainability. This study is significant to Batch Flow because the system promotes organized and efficient bakery management processes.

Meanwhile, Githa and Raharja (2021) stated that electronic supply chain management systems improve business efficiency in bakery operations. Likewise, the researchers found that e-SCM technologies strengthen coordination between suppliers, businesses, and customers. This literature supports Batch Flow because the proposed system integrates supply chain activities into a web-based platform.

Suryawan and Nurcaya (2021), emphasized proper raw material supply management reduces total inventory costs in bakery businesses. Therefore, methods such as Economic Order Quantity (EOQ) improve inventory efficiency and stock replenishment decisions. This literature is related to Batch Flow because the proposed system aims to automate inventory tracking and supply management.

Moreover, Patil et al. (2023) said that inventory management for perishable products is essential in bakery industries because bakery items have a short shelf life. Meanwhile, the study emphasized that inventory optimization helps reduce product wastage and maintain product freshness. This supports Batch Flow because the system is designed to monitor bakery inventory and minimize spoilage.

Furthermore, Medeiros, Medeiros, Lima, and Silva (2024) explained that bakery industries benefit from inventory management systems that use continuous review models and Economic Order Quantity techniques. Subsequently, the study revealed that proper inventory systems reduce operational expenses and improve stock availability. This literature is relevant because Batch Flow also focuses on efficient inventory control and stock monitoring.

Aljohani (2023), explained that optimized bakery distribution networks reduce travel distance and minimize food waste. Hence, efficient supply chain strategies improve product delivery and customer satisfaction in bakery businesses. This study is connected to Batch Flow because the proposed system aims to improve coordination of bakery supplies and product distribution.

According to Fernandez-Carames, Blanco-Novoa, Froiz-Miguez, and Fraga-Lamas (2024), automated inventory and traceability systems improve supply chain efficiency through real-time monitoring technologies. Finally, the researchers emphasized that automation strengthens accuracy, traceability, and inventory control in modern supply chain management. This literature supports Batch Flow because the proposed system also focuses on automated monitoring and efficient inventory management in bakery operations.

Related Systems

This chapter presents the related systems relevant to the proposed web-based bakery management project entitled Batch Flow: A POS Based Supply Chain Management System. The reviewed systems provide insights into existing technologies and platforms used in point-of-sale operations, inventory monitoring, supply chain management, sales tracking, and customer management within bakery and food service businesses. Moreover, these systems were examined to identify their features, functionalities, strengths, and weaknesses that may contribute to the development of the proposed system.

Toast POS was developed by Toast, Inc. as a cloud-based point-of-sale system specifically designed for food service businesses such as bakeries, cafés, and restaurants. The system offers features including inventory tracking, sales reporting, employee management, customer order monitoring, and online ordering integration. Furthermore, Toast POS helps bakery owners manage daily operations efficiently by providing real-time data analytics and automated stock monitoring. The system's strength lies in its user-friendly interface, mobile accessibility, and ability to integrate with payment processing systems. However, the software heavily depends on internet connectivity, which may disrupt operations during connection failures. In relation to Batch Flow: A POS Based Supply Chain Management System, Toast POS serves as a reference for integrating sales monitoring and inventory management into a single platform to improve bakery operational efficiency.

Shopify POS was developed by Shopify Inc. to combine physical store operations with e-commerce management. The system enables bakery businesses to synchronize online and in-store sales, monitor inventory levels, and manage customer transactions through a centralized dashboard. Additionally, Shopify POS supports multiple payment methods and customer loyalty programs that enhance user experience and business performance. One of its strengths is its scalability, making it suitable for both small and expanding bakery businesses. However, advanced features and third-party integrations often require additional subscription fees, which may increase operational expenses. This system is related to Batch Flow because it demonstrates how centralized inventory and sales synchronization can improve bakery supply chain management and customer service efficiency.

Square for Restaurants was developed by Block, Inc. as a restaurant-focused point-of-sale system that supports sales transactions, inventory management, and employee scheduling. The platform provides bakeries with tools for digital receipts, order customization, sales analytics, and inventory monitoring. Moreover, its simple interface and affordable pricing make it highly accessible to small and medium-sized bakery businesses. The system's strength lies in its ease of setup and efficient payment processing capabilities. However, its customization options may be limited compared to enterprise-level POS systems. This related system contributes to the development of Batch Flow by providing insights into efficient sales processing and inventory tracking functionalities that can be adapted for bakery operations.

Oracle MICROS Simphony was developed by Oracle Corporation as an enterprise-level management platform for restaurants and food establishments. The system integrates POS operations, inventory management, kitchen monitoring, and supply chain processes into one centralized solution. Furthermore, it supports multi-branch management and cloud-based reporting, allowing businesses to monitor operations in real time. Its primary strength is its comprehensive functionality and advanced analytics capabilities for large-scale operations. However, the system may be expensive and technically complex for small bakery businesses to implement. This system is relevant to Batch Flow because it demonstrates how integrated management solutions can streamline bakery operations, inventory control, and supply chain processes effectively.

Loyverse POS was developed by Loyverse as a free cloud-based POS solution intended for small retail stores, cafés, and bakery businesses. The system includes features such as inventory management, customer loyalty programs, employee monitoring, and sales reporting. Additionally, it allows bakery owners to access business information through mobile devices for real-time operational monitoring. One of its strengths is its affordability and accessibility for startup businesses with limited budgets. However, advanced features and system integrations may require premium subscriptions. This related system is connected to Batch Flow because it highlights the importance of affordable and accessible digital management solutions for improving bakery operations and inventory control.

Lightspeed Retail POS was developed by Lightspeed Commerce Inc. as a cloud-based retail and inventory management platform. The system enables bakery businesses to automate inventory tracking, supplier management, and sales reporting through an integrated interface. Furthermore, it provides analytics and forecasting tools that assist business owners in making informed operational decisions. Its strength lies in its advanced reporting capabilities and scalability for growing businesses. However, the software subscription and additional features may be costly for smaller bakery enterprises. The system is relevant to Batch Flow because it demonstrates how automated inventory management and supplier coordination can enhance bakery supply chain efficiency.

Revel Systems POS was developed by Revel Systems as an iPad-based point-of-sale and business management solution for food establishments. The system integrates inventory management, customer relationship management, sales tracking, and employee management into one platform. Additionally, it supports real-time reporting and multi-location business operations, making it suitable for expanding bakery businesses. One of its strengths is its flexibility and ability to handle high transaction volumes efficiently. However, the system requires compatible hardware and may involve high setup costs. This related system provides Batch Flow with ideas on integrating centralized sales and inventory management for improved bakery operational performance.

SAP Business One was developed by SAP SE as an enterprise resource planning (ERP) system for small and medium-sized businesses. The software supports inventory management, procurement, accounting, customer management, and supply chain operations within one integrated platform. Furthermore, it provides real-time analytics and automated reporting to improve operational decision-making. Its major strength is its comprehensive business process integration and scalability. However, implementation and maintenance may require technical expertise and employee training. This system is related to Batch Flow because it demonstrates the value of integrating bakery supply chain processes, sales management, and inventory monitoring into a unified system.

Zoho Inventory was developed by Zoho Corporation as a cloud-based inventory and order management system suitable for retail and food-related businesses. The software provides features such as warehouse monitoring, order tracking, inventory synchronization, and supply chain automation. Furthermore, it allows businesses to monitor stock movement and sales performance across multiple channels. Its strength lies in its affordability, automation tools, and integration capabilities with other business applications. However, the free plan offers limited functionalities that may not fully support larger bakery operations. This system is relevant to Batch Flow because it highlights the importance of automated inventory tracking and order management in improving bakery supply chain efficiency and reducing stock-related issues.

KORONA POS was developed by COMBASE USA, is a retail point-of-sale system widely utilized by small to medium-scale artisanal bakeries due to its deep stock management parameters. The software features automated low-stock warnings, real-time inventory adjustments during checkout, and an integrated supplier interface that automates the generation of purchase requests once specific ingredient thresholds are reached. It also supports centralized data tracking via a back-office cloud dashboard, allowing multi-branch bakery managers to monitor waste data and stock movement concurrently. KORONA POS is fundamentally linked to the conceptual goals of Batch Flow because it underscores the importance of automated supplier notification loops and centralized inventory visibility in modern commercial baking environments.

CHAPTER 3

SYSTEM ANALYSIS AND DESIGN

Development Methodology

The development of Batch Flow: A POS Based Supply Chain Management System utilized the Agile Software Development Methodology. Agile is a flexible and iterative approach that focuses on continuous improvement, collaboration, and rapid delivery of system features. This methodology was selected because it allows the developers to adapt to changing requirements and continuously improve the system based on user feedback. Furthermore, Agile supports incremental development, enabling the researchers to divide the system into smaller modules such as inventory management, sales monitoring, supply chain tracking, and point-of-sale operations. Through regular testing and evaluation, the developers can immediately identify and resolve issues to ensure the quality and reliability of the system.

Moreover, the Agile methodology promotes close communication between developers, bakery staff, and system users throughout the development process. Each phase of development involves planning, designing, coding, testing, and reviewing to ensure that the system meets the operational needs of the bakery business. Additionally, Agile allows the researchers to implement updates and additional features without significantly affecting the entire system structure. This approach is beneficial for Batch Flow because bakery operations may require continuous adjustments in inventory processes, sales transactions, and supply chain monitoring. Consequently, Agile provides an efficient and user-centered framework for developing a reliable, scalable, and responsive web-based supply chain management system for bakery businesses.

In addition, the Agile Software Development Methodology enhances the efficiency and productivity of the development team by encouraging regular collaboration and continuous feedback during each sprint cycle. Through iterative development, the researchers can prioritize important system functions and deliver working components within a shorter period of time. This process enables the bakery management and staff to evaluate the usability and performance of the system early in the development stage, allowing necessary modifications to be implemented immediately. As a result, the system becomes more aligned with the actual business processes and operational requirements of the bakery. Likewise, Agile supports risk management by identifying technical and functional issues at an early stage, thereby minimizing delays and reducing development costs. Therefore, the methodology contributes significantly to the successful creation of Batch Flow: A POS Based Supply Chain Management System by ensuring adaptability, system quality, and continuous improvement throughout the project development lifecycle.

Figure 1. Agile Methodology for Batch Flow

Phase 1: Requirements

The Requirements phase is the initial stage of the Agile Methodology where the developers gather all necessary information about the system to be developed. In this phase, the researchers identify the needs and problems encountered by bakery businesses in managing sales, inventory, suppliers, and supply chain operations. Furthermore, interviews, surveys, and observations are conducted to determine the system requirements and expected functionalities of Batch Flow: A POS Based Supply Chain Management System.

Additionally, this phase helps define the scope, objectives, and limitations of the proposed system. The developers analyze user expectations and prioritize important features such as point-of-sale transactions, inventory tracking, sales reporting, and supplier management. Through proper requirement gathering, the researchers can ensure that the system will effectively address the operational challenges faced by bakery businesses.

Phase 2: Design

The Design phase focuses on planning the structure, interface, and functionality of the system before actual development begins. During this stage, the researchers create system models, database structures, wireframes, and interface layouts for the Batch Flow system. Moreover, the developers determine how the different modules, such as inventory management, sales processing, and supply chain monitoring, will interact with one another.

Furthermore, this phase ensures that the system design is user-friendly, organized, and efficient for bakery staff and administrators. The researchers also plan the navigation flow, security measures, and overall appearance of the web system to improve usability and accessibility. Through proper system design, the developers can minimize errors and establish a strong foundation for the development process.

Phase 3: Development

The Development phase is the stage where the actual coding and programming of the system are performed. In this phase, the developers build the features and functionalities identified during the requirements and design stages. Additionally, the researchers create modules for point-of-sale operations, inventory management, supplier tracking, customer transactions, and report generation for the Batch Flow system.

Moreover, Agile development follows an iterative process where the system is developed in smaller parts called sprints. Each sprint focuses on completing specific features that can be tested and improved continuously. This approach allows the developers to identify problems early and make necessary adjustments without affecting the entire system. As a result, the development process becomes more flexible, efficient, and adaptable to user feedback.

Phase 4: Testing

The Testing phase involves evaluating the functionality, performance, and reliability of the system to identify errors or issues. During this stage, the developers conduct different testing procedures such as functionality testing, usability testing, and system integration testing to ensure that Batch Flow operates properly. Furthermore, the researchers verify whether all modules and features meet the specified system requirements and user expectations.

Additionally, testing helps improve the overall quality and security of the web system before deployment. Errors, bugs, and technical issues discovered during testing are immediately corrected to prevent operational problems in actual bakery use. This phase is important because it ensures that the system provides accurate transactions, reliable inventory tracking, and smooth supply chain management processes.

Phase 5: Deployment

The Deployment phase is the process of launching and implementing the completed system for actual use. In this phase, the developers install the Batch Flow system into the bakery's operational environment and ensure that all features are functioning properly. Furthermore, user training and orientation are conducted to help bakery staff understand how to operate the system effectively.

Moreover, deployment allows the researchers to monitor the real-time performance of the system and evaluate its effectiveness in improving bakery operations. Necessary configurations and adjustments may also be performed to ensure compatibility with the bakery's workflow and hardware devices. Through proper deployment, the system becomes accessible and ready to support daily business activities efficiently.

Phase 6: Review and Feedback

The Review and Feedback phase is the final stage of the Agile Methodology where the developers collect comments, suggestions, and evaluations from users regarding the system. In this phase, bakery staff and administrators assess the effectiveness, usability, and performance of the Batch Flow system. Furthermore, the researchers analyze user feedback to identify strengths, weaknesses, and areas that require improvement.

Additionally, this phase supports continuous system enhancement and maintenance. The developers use the gathered feedback to implement updates, fix issues, and improve system functionalities in future development cycles. Agile emphasizes continuous improvement, making the system more responsive to changing bakery business needs and operational requirements over time.

Proposed System Features

A.Operational Features

- **Point-of-Sale (POS) Management.** This feature allows bakery staff to process customer transactions quickly and accurately. It records purchased products, calculates totals automatically, generates digital receipts, and updates sales data in real time. This feature minimizes manual computation errors and improves customer service efficiency during peak business hours. Additionally, the POS module stores transaction histories that can be reviewed by the administrator for monitoring daily sales performance. It also supports faster checkout processes, helping the bakery provide a smoother and more organized customer experience.

- **Inventory Management.** In this feature they monitors the quantity of raw materials and finished bakery products available in stock. It automatically updates inventory levels whenever products are sold or new supplies are added, helping staff maintain accurate inventory records at all times.
  This feature also prevents overstocking and shortages by providing alerts when supplies are running low. Through automated inventory tracking, the bakery can reduce waste, improve stock organization, and ensure that products remain available for customers.
- **Supply Chain Management.** On this feature it will feature helps manage supplier information, deliveries, and restocking activities. It records supplier details, tracks incoming orders, and monitors delivery schedules to ensure that bakery ingredients and materials arrive on time. Moreover, this feature improves coordination between the bakery and suppliers by organizing procurement processes efficiently. It helps maintain a continuous supply of ingredients, reducing delays in production and ensuring consistent bakery operations.
- **Product Management.** In this feature enables administrators to add, edit, update, and remove bakery products within the system. Product details such as name, category, price, quantity, and description can be managed easily through the web interface**.** This feature ensures that product information displayed in the POS system remains accurate and updated. It also allows bakery owners to introduce seasonal products, promotional items, or price adjustments without difficulty.
- **Sales Reporting and Analytics.** This feature generates reports regarding daily, weekly, or monthly sales transactions. These reports help administrators analyze product performance, identify best-selling items, and monitor overall business profitability. In addition, the system presents organized data that supports better business decision-making. By reviewing sales trends and customer purchasing behavior, bakery owners can improve marketing strategies and inventory planning.

B.MIS Features

- **Management Reporting System.** In this feature it provides organized and detailed reports about sales, inventory, expenses, and supply chain activities. It helps bakery managers monitor business performance through daily, weekly, and monthly summaries that are automatically generated by the system. This feature supports decision-making by presenting accurate and updated information in a clear format. Managers can use these reports to evaluate operations, identify problems, and create effective strategies for improving bakery performance and profitability.
- **Decision Support Feature.** This feature assists administrators in making informed business decisions by analyzing sales trends, inventory levels, and customer demand. The system gathers operational data and transforms it into useful insights that can guide management actions. Through this feature, bakery owners can determine which products sell the most, identify slow-moving items, and decide when to restock supplies. It improves planning and helps reduce losses caused by poor inventory or production decisions.
- **Real-Time Data Monitoring.** On this side of the feature it allows the managers to view current business activities instantly through the web system dashboard. Information such as ongoing sales transactions, available stock levels, and supply updates are displayed in real time. This feature enables faster responses to operational issues and helps management maintain smooth bakery operations. By accessing updated information immediately, managers can avoid delays, shortages, and inaccurate reporting.
- **Inventory Information System.** In this features it records and organizes all inventory-related data, including raw materials, bakery products, and stock movement. It provides accurate monitoring of inventory usage and availability within the bakery. This feature helps management maintain proper stock levels and reduce wastage caused by expired or excess ingredients. It also ensures that inventory records remain consistent and easily accessible for auditing and operational purposes.
- **Customer Sales Tracking.** The feature of this is to stores transaction histories and customer purchasing information within the system. It helps bakery managers analyze customer buying behavior, preferred products, and sales frequency. By understanding customer preferences, the bakery can improve product offerings, create promotional strategies, and enhance customer satisfaction. This feature also supports business growth by helping management identify profitable products and sales opportunities.

- **User Management and Security.** This feature controls the system access by assigning usernames, passwords, and user roles to authorized personnel. Different access levels are provided to administrators, cashiers, and staff to ensure proper system usage. This feature protects important business information from unauthorized access and data manipulation. It also promotes accountability because system activities can be tracked based on the logged-in user performing each operation.

System Overview Diagram

This section presents the System Overview Diagram of the proposed web-based bakery system, Batch Flow: A POS Based Supply Chain Management System. It illustrates the overall structure of the system, including the interaction between users, processes, and database components. Moreover, the diagram provides a clear visualization of how information flows within the system to support sales, inventory, supply chain, and transaction management operations.

Use Case Diagram

The Use Case Diagram illustrates the functional interactions between five primary actors: Admin, Cashier, Customer, Supplier, and Manager. The system is designed for in-store operations, where the Admin handles backend configurations like user management, pricing, and system integrity, while the Cashier manages the direct sales process, including transaction creation, receipt generation, and shift logging. The Customer interacts with the system by browsing products, returning goods, and participating in a loyalty program that shares a relationship with the cashier's processing activities. On the supply side, the Supplier fulfills purchase orders and delivers goods, which feeds into the inventory management monitored by the Manager, who also analyzes sales data and generates comprehensive reports such as profit and audit trails. In addition, the Use Case Diagram demonstrates how each actor contributes to the overall workflow and operational efficiency of the bakery business. The interactions among the users and the system ensure that transactions, inventory monitoring, customer management, and supply chain activities are properly coordinated in real time. Through this integration, the system minimizes manual processes, reduces human errors, and improves the accuracy of business records. Furthermore, the diagram highlights the connection between operational activities and managerial decision-making by allowing reports and transaction data to be instantly accessible. As a result, the proposed system supports faster service, better inventory control, improved customer satisfaction, and more effective business management within the bakery environment.

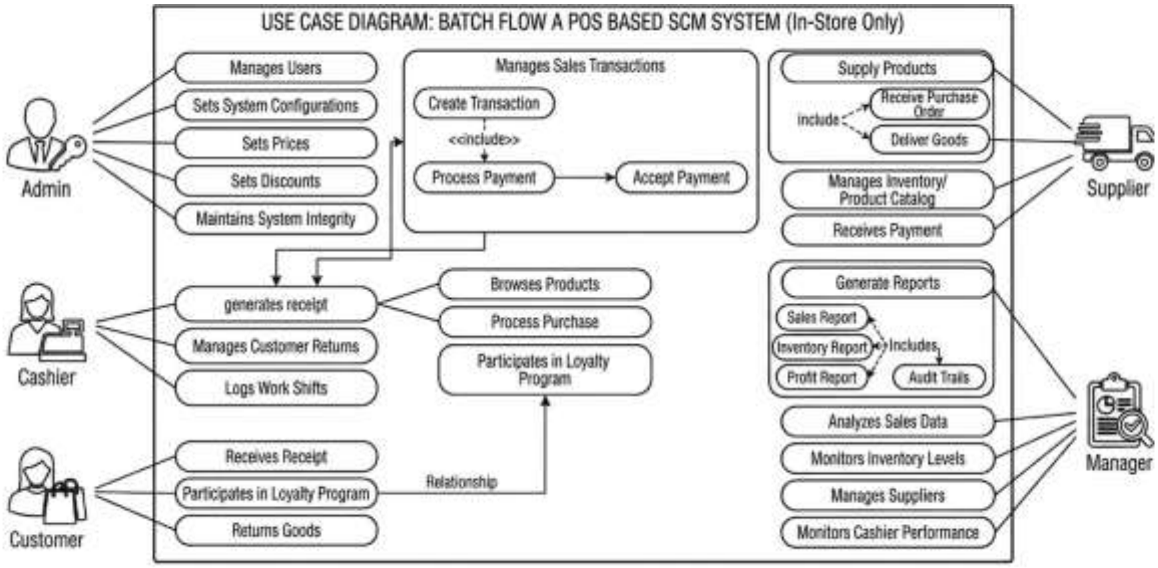

Figure 2. Use Case Diagram

Data Flow Diagram (DFD)

DFD Level 0

The Level 0 DFD provides a high-level overview of the system's boundaries, showing the central "Batch Flow" process interacting with five external entities: Customer, Supplier, Admin, Cashier, and Manager. In this stage, the Customer sends purchase orders and receives invoices, while the Supplier handles deliveries and receives payments. The Admin provides master data (items and categories) and receives reports, the Cashier exchanges sales data for transaction details, and the Manager issues decisions or requests to receive analytical reports. This diagram serves as the foundation for understanding how information flows throughout the entire system and how each external entity communicates with the main process. It highlights the major inputs and outputs involved in daily bakery operations, ensuring that all critical business activities are properly integrated within a single system. By presenting the overall data movement in a simplified manner, the diagram helps stakeholders easily identify the system's scope, functional relationships, and the flow of information necessary to support efficient decision-making and streamlined business operations.

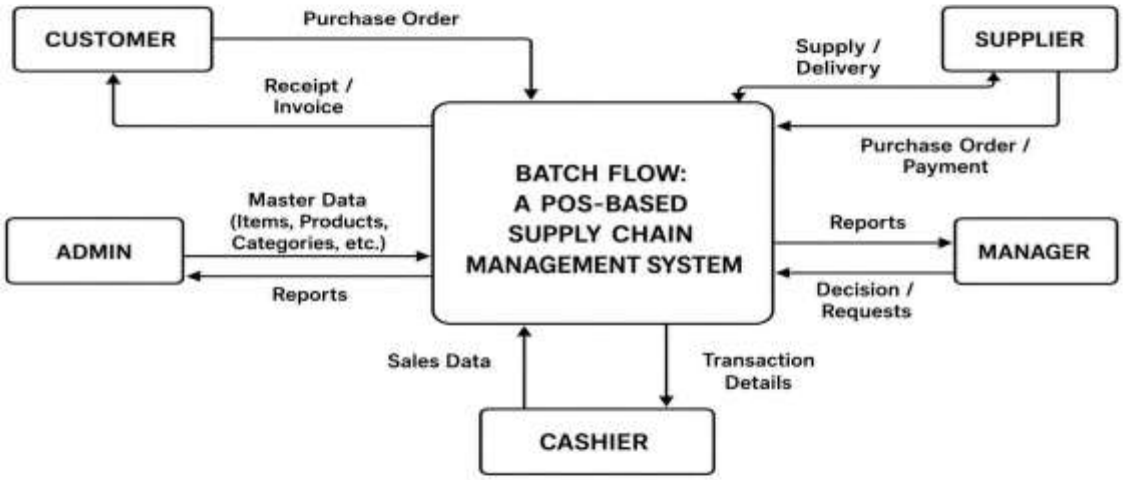

Figure 3. DFD Level 0

DFD Level 1

The Level 1 DFD decomposes the main system into five primary functional areas: Manage Sales (POS), Manage Inventory, Manage Purchases, Manage Suppliers, and Generate Reports. This level introduces six specific data stores (Sales, Products, Inventory, Purchases, Suppliers, and Reports) that allow information to be persisted across different operations. For instance, "Manage Sales" interacts with the Sales and Products data stores to process customer orders, while "Manage Purchases" handles admin requests by pulling from the Purchases and Suppliers data stores.

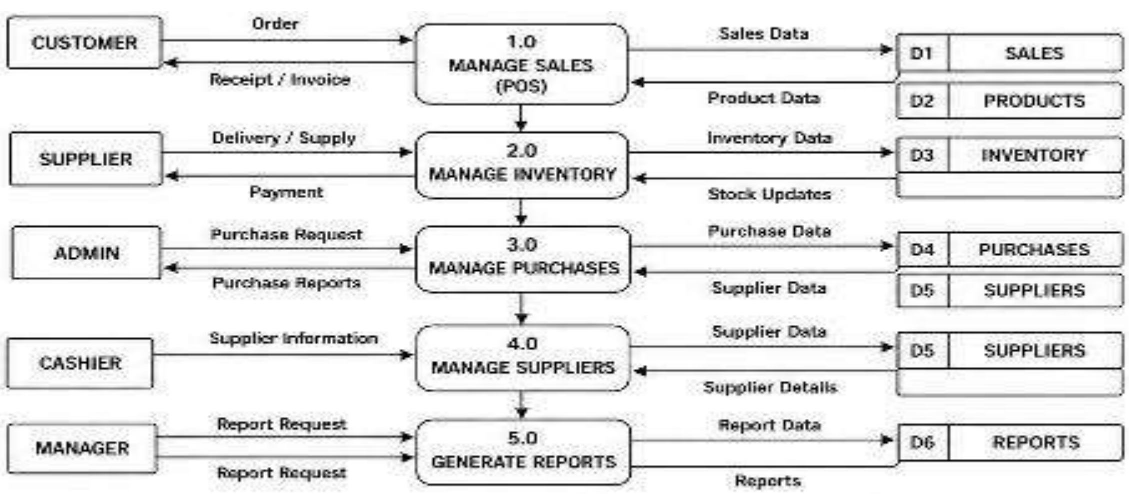

Figure 4. DFD Level 1

DFD Level 2

The Level 2 DFD focuses specifically on the granular steps of the sales lifecycle, breaking the POS process down into five sub-processes: Input Sales (1.1), Check Inventory (1.2), Process Payment (1.3), Record Sale (1.4), and Generate Receipt (1.5). It tracks the flow from the moment a customer provides order items through inventory verification and payment status updates until the final sale is recorded in the Sales and Inventory data stores. The cycle concludes with the generation of a receipt or invoice, which is then delivered back to the customer.

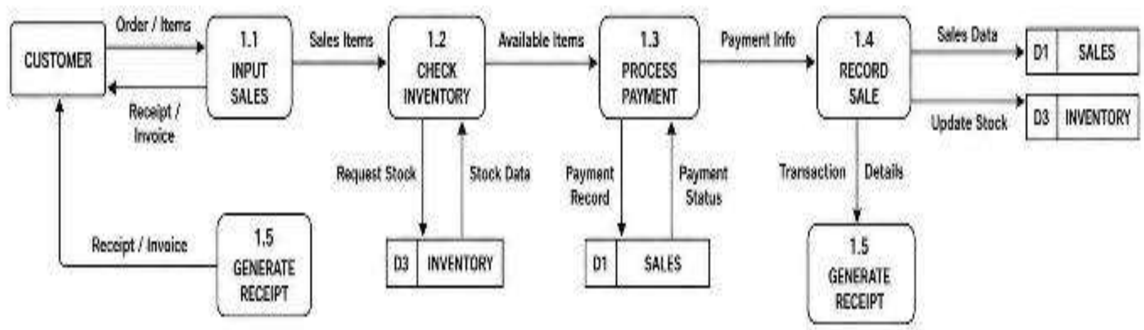

Figure 5. DFD Level 2

System Flowchart

The system flowchart for Batch Flow: A POS-Based Supply Chain Management System illustrates the overall process and user interaction within the system. The process begins with the Start symbol, where users access the system through the User Login module. The entered credentials are checked in the Validate Credentials process, and if the information is incorrect, the system redirects the user to the Invalid Credentials process. Once the login is verified, the system determines the User Role, which branches into five different user categories: Admin, Manager, Cashier, Supplier, and Customer. Each user role has specific functions and responsibilities within the system. The Admin manages users, products, suppliers, system settings, and reports. The Manager oversees dashboards, sales, inventory, purchase reports, and approvals. The Cashier handles POS transactions, payments, receipts, and stock checking. The Supplier manages purchase orders, deliveries, invoices, and payment receipts, while the Customer browses products, purchases items, receives receipts, requests returns, and provides feedback. After completing their assigned tasks, all users proceed to the Logout process, which finally leads to the End of the system flow.

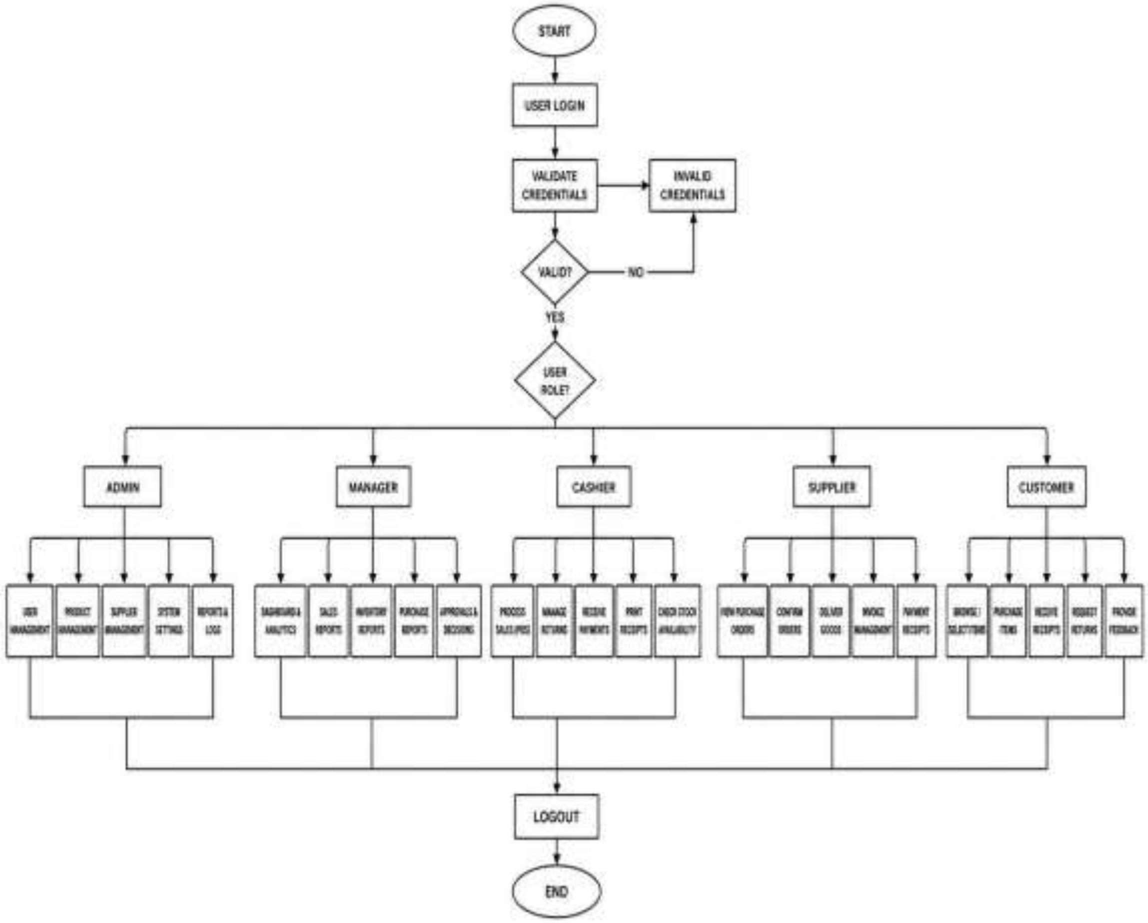

Figure 6. System Flowchart

Admin Flowchart

The Admin flowchart for the Batch Flow: A POS-Based Supply Chain Management System illustrates the step-by-step process performed by the administrator within the system. The process begins with the Admin Login, where the administrator enters a username and password. The system then performs Credential Validation to verify the entered information. If the credentials are invalid, the process redirects to the Invalid Credentials step and prompts the user to try again. Once validated, the administrator gains access to the Admin Dashboard, where different management modules are available. These modules include User Management, Product Management, Supplier Management, System Settings, and Reports & Logs. The administrator can manage records, update system configurations, monitor reports, and maintain system data. After completing all necessary tasks, the admin saves the updates, logs out of the system, and the process ends.

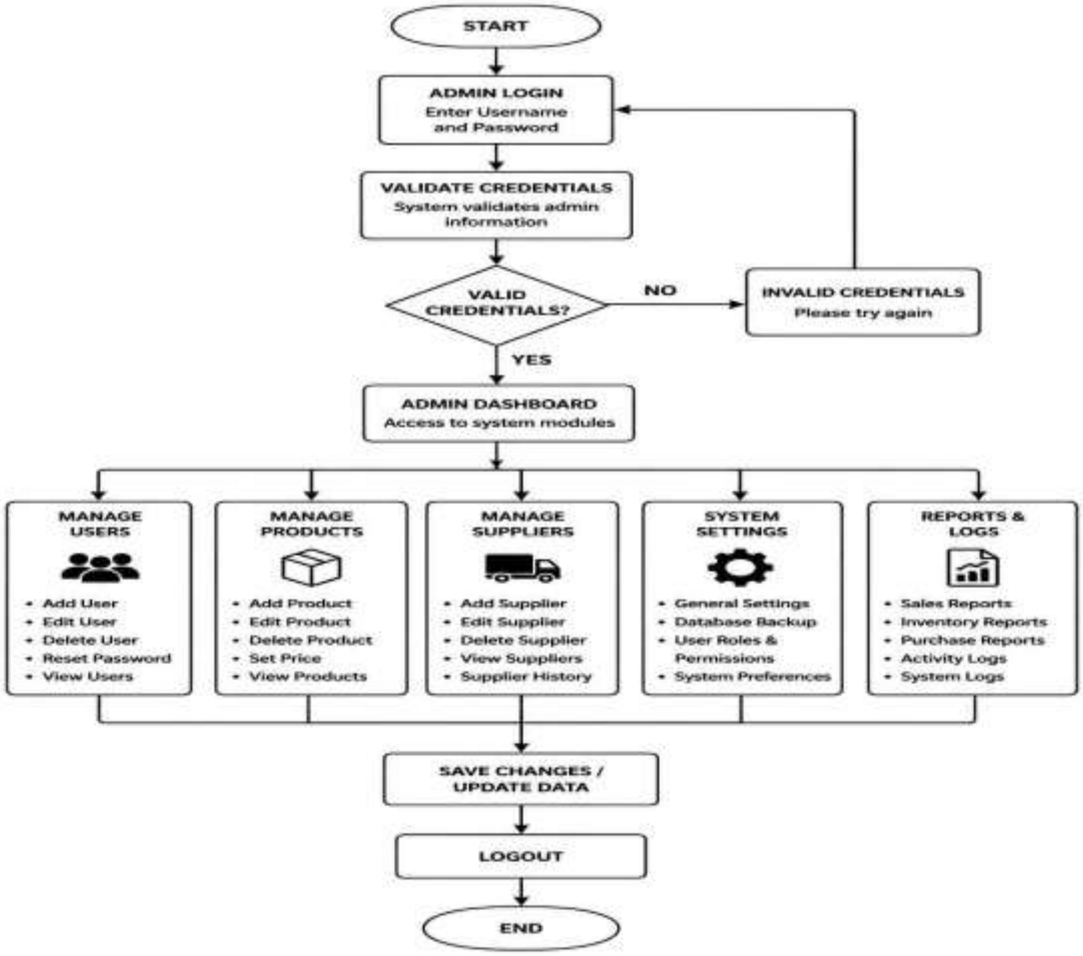

Figure 7. Admin Flowchart Continuous

Cashier Flowchart

This cashier flowchart outlines the operational flow for a point-of-sale system, beginning with the cashier's login. The system first validates the credentials; if they are incorrect, the user is prompted to try again. Upon a successful login, the main Cashier Dashboard is accessed, allowing the user to select from five major operational modules: processing a sale, managing the payment details, printing a receipt, checking current stock, or processing a return. Once a specific task within these modules is completed, the system saves the entire transaction data.

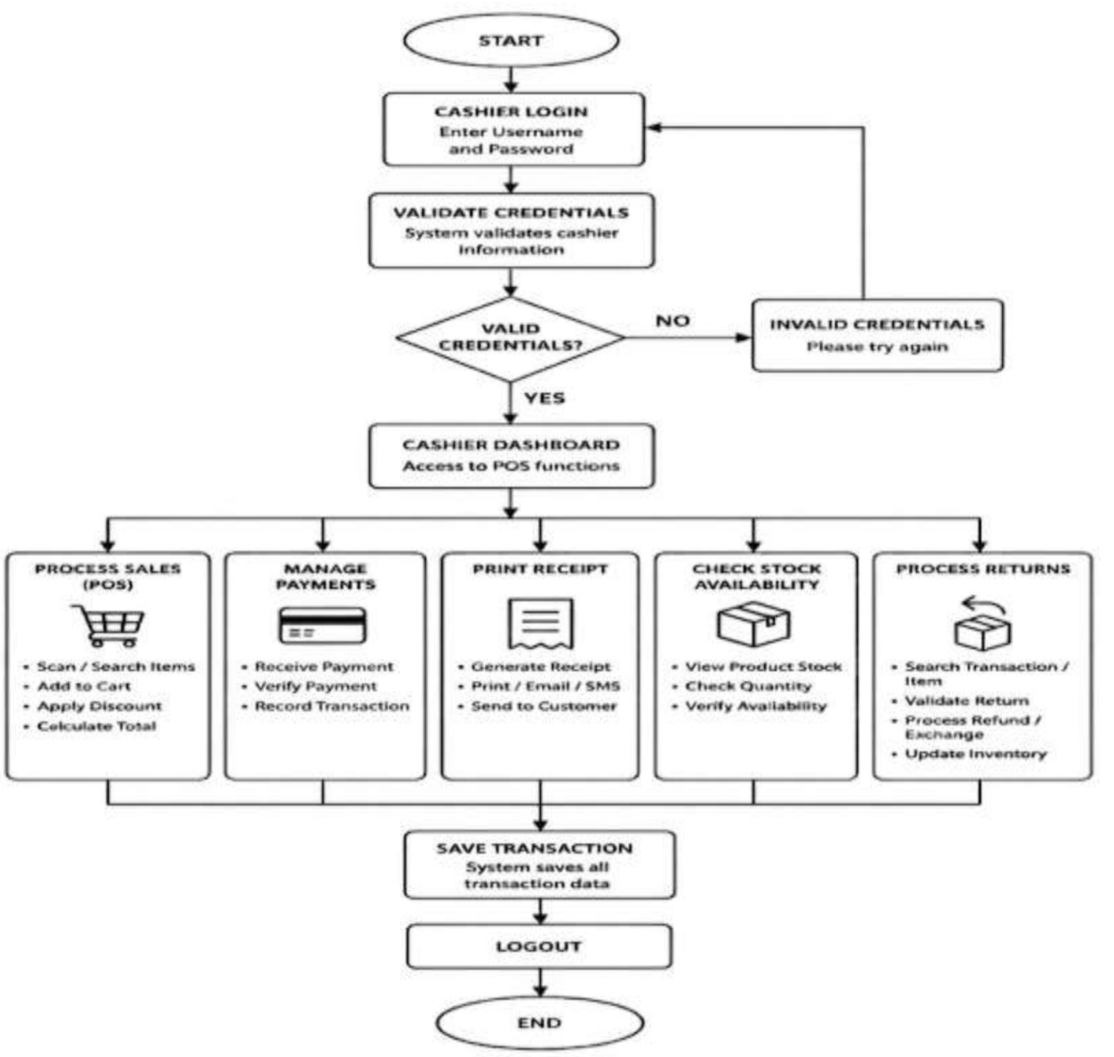

Figure 8. Cashier Flowchart Continuous

Customer Flowchart

The Customer Flowchart illustrates a comprehensive user journey within a retail or e-commerce system. It begins with a secure login and role validation, after which the customer enters a cycle of browsing products and managing their virtual cart. The process moves into a critical decision phase where the user reviews their items, applies discounts, and proceeds to a multi-step payment gateway. Once the transaction is confirmed, the system automatically triggers a digital receipt and inventory update, while offering the customer optional post-purchase actions such as providing feedback or requesting a return. The journey concludes with a logout, ensuring all transaction data is secured before the session ends.

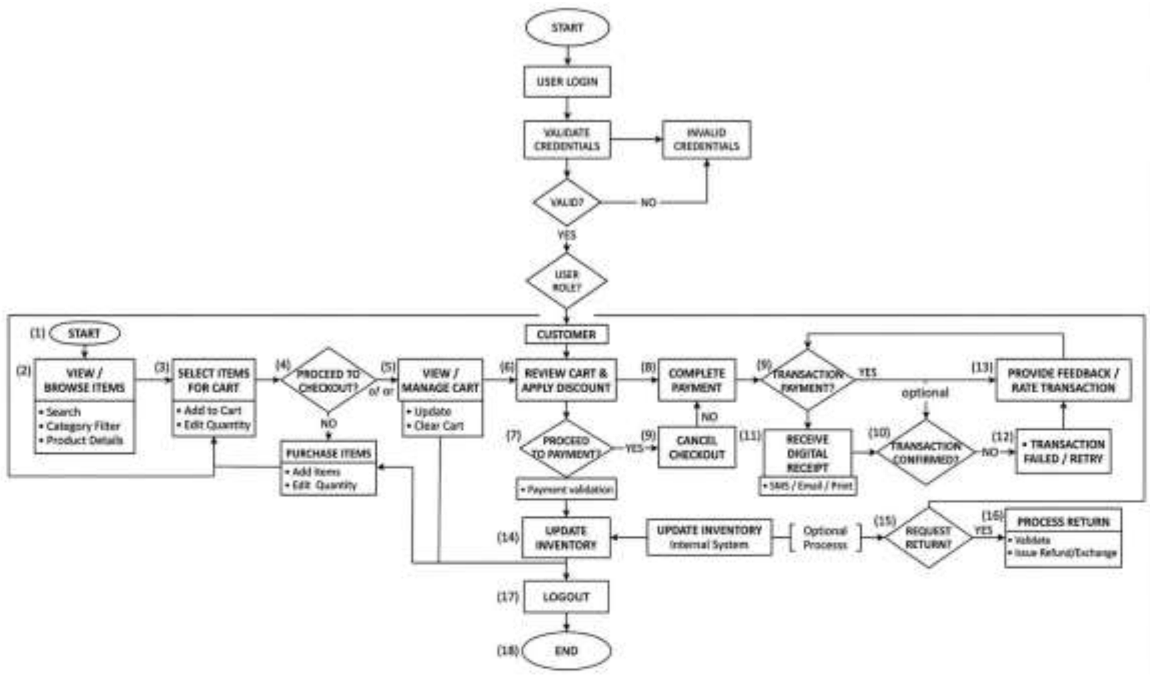

Figure 9. Customer Flowchart Continuous

Supplier Workflow

The Supplier Workflow details the business-to-business interaction between a supplier and the core system. After a secure authentication process, the supplier gains access to a specialized dashboard focused on the supply chain lifecycle. This involves viewing incoming purchase orders, managing production schedules, and checking inventory levels to ensure fulfillment capability. The flow includes critical decision points for order approval and stock availability; if stock is sufficient, the system moves to quality control and packing, followed by the generation of invoices and shipping labels. The process concludes by updating the inventory status and finalizing the transaction data before the supplier logs out, ensuring a seamless link between production and delivery. The Supplier Workflow highlights the importance of accurate coordination and communication between suppliers and the bakery management system. By automating purchase order tracking, stock verification, and delivery updates, the system reduces delays, minimizes manual errors, and improves the efficiency of procurement operations. The inclusion of quality control procedures ensures that delivered products meet the required standards before shipment, which helps maintain product consistency and customer satisfaction.

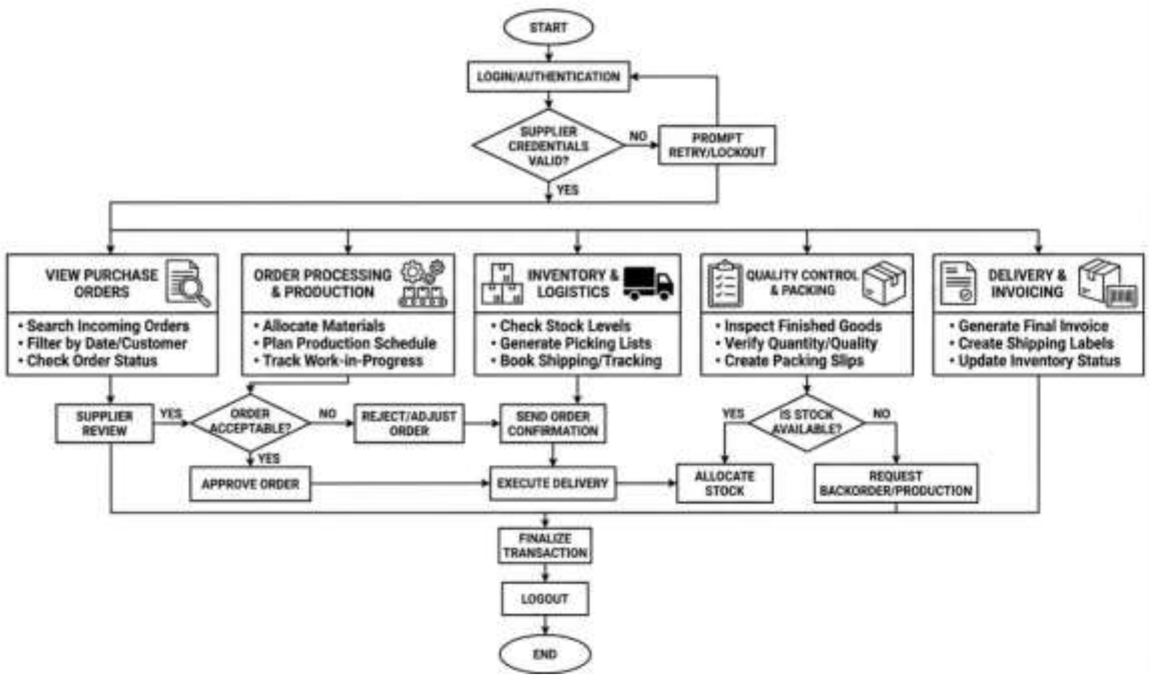

Figure 10. Supplier Workflow Continuous

Manager Flowchart

The Manager flowchart illustrates the administrative and strategic oversight within the system. After a secure login and role verification, the manager gains access to a comprehensive dashboard divided into three core pillars that includes Analytics & Reporting, Operational Oversight, and Strategic Planning. This allows the manager to review performance data, monitor daily staff activities, and handle operational issues with corrective actions. Furthermore, the flow includes high-level decision-making processes, such as developing purchase forecasts and approving budgets. All these activities converge into regular team review meetings to ensure organizational alignment, concluding with a secure logout to protect sensitive management data. In addition, the Manager flowchart demonstrates how the system supports effective leadership and informed decision-making within the bakery business. By centralizing operational reports, sales analytics, inventory status, and employee performance monitoring into one platform, the manager can quickly identify business trends, operational inefficiencies, and areas that require improvement. The inclusion of strategic planning and budget approval processes also ensures that resources are allocated efficiently to maintain profitability and sustainable growth.

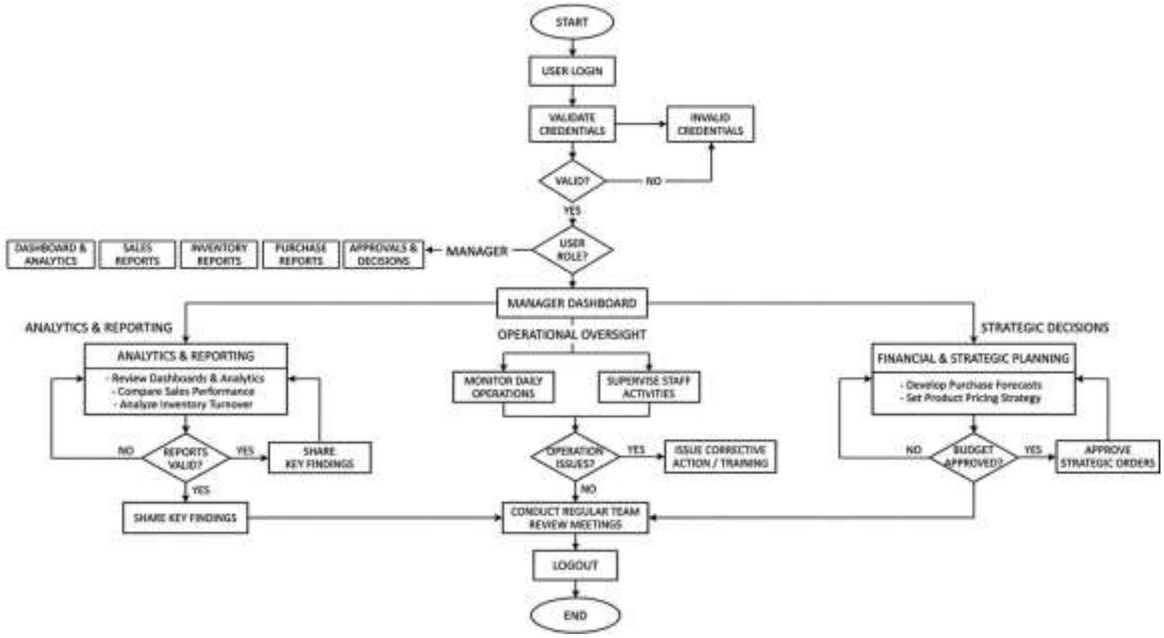

Figure 11. Manager Flowchart Continuous

Entity Relationship Diagram (ERD)

The Entity Relationship Diagram (ERD) of Batch Flow: A POS Based Supply Chain Management System illustrates how the different entities within the bakery management system are connected to support daily business operations. The system is centered on the Sales entity, which records all customer transactions processed by employees or cashiers. Each sale is associated with the Users table through the user_id, allowing the system to identify which employee handled the transaction. The Sale_Items entity serves as a bridge between sales and products by storing detailed information about purchased items, including quantity, price, and subtotal. Meanwhile, the Products table contains essential product information such as product name, category, pricing, stock quantity, and reorder level, enabling efficient inventory monitoring and product management. The relationship between Products and Categories allows bakery items to be grouped according to their type, making product organization more systematic and easier to manage. In addition, the ERD demonstrates the supply chain and inventory management capabilities of the system. The Suppliers entity stores supplier details such as company name, contact person, and address, which are connected to the Deliveries table to track incoming product shipments. Each delivery contains multiple records in the Delivery_Items table, where the delivered products, quantities, and unit costs are recorded. Furthermore, the Inventory_Logs entity monitors all inventory movements, including stock additions and deductions, ensuring accurate stock tracking and transparency within the bakery operations. The Expenses table is also linked to the Users entity to record operational expenses made by employees, including utility costs, maintenance, and other business-related expenditures. Overall, the ERD presents a well-structured database design that integrates point-of-sale operations, inventory management, supplier coordination, delivery tracking, and expense monitoring into one centralized bakery management system.

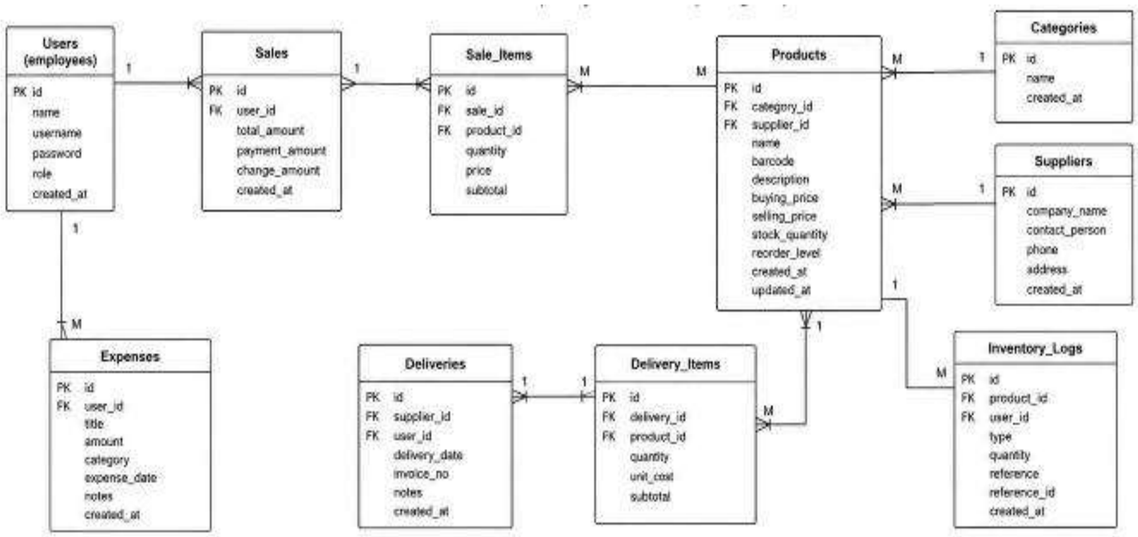

Figure 12. Entity Relationship Diagram (ERD)

Database Design

The database design for Batch Flow that was implemented in MySQL, The tables, their fields, data types, and key relationships are described below.

Table: users

users		
id	INTEGER	PK
name	TEXT	
username	TEXT	UNIQUE
password	TEXT	
role	TEXT	
created_at	DATETIME	

Table: sales

sales		
id	INTEGER	PK
user_id	INTEGER	FK
total_amount	REAL	
payment_amount	REAL	
change_amount	REAL	
created_at	DATETIME	

Table: sale_items

sale_items		
id	INTEGER	PK
sale_id	INTEGER	FK
product_id	INTEGER	FK
quantity	INTEGER	
price	REAL	
subtotal	REAL	

Table: products

products		
id	INTEGER	PK
category_id	INTEGER	FK
supplier_id	INTEGER	FK
name	TEXT	
barcode	TEXT UNIQUE	
description	TEXT	
buying_price	REAL	
selling_price	REAL	
stock_quantity	INTEGER	
reorder_level	INTEGER	
created_at	DATETIME	
updated_at	DATETIME	

Table: categories

categories	
id	INTEGER
name	TEXT
created_at	DATETIME

Table: suppliers

suppliers		
id	INTEGER	PK
company_name	TEXT	
contact_person	TEXT	
phone	TEXT	
address	TEXT	
created_at	DATETIME	

Table: expenses

expenses		
id	INTEGER	PK
user_id	INTEGER	FK
title	TEXT	
amount	REAL	
category	TEXT	
expense_date	DATE	
notes	TEXT	
created_at	DATETIME	

Table: deliveries

deliveries		
id	INTEGER	
supplier_id	INTEGER	FK
user_id	INTEGER	FK
delivery_date	DATE	
invoice_no	TEXT	
notes	TEXT	
created_at	DATETIME	

Table: delivery_items

delivery_items		
id	INTEGER	
delivery_id	INTEGER	FK
product_id	INTEGER	FK
quantity	INTEGER	
unit_cost	REAL	
subtotal	REAL	

Table: inventory_logs

inventory_logs		
id	INTEGER	
product_id	INTEGER	FK
user_id	INTEGER	FK
type	TEXT	
quantity	INTEGER	
reference	TEXT	
reference_id	INTEGER	
created_at	DATETIME	

Functional Requirements

Functional requirements describe the specific features and functions that the system must perform to support bakery operations, sales transactions, inventory control, and supply chain management. These requirements ensure that the system can meet the daily operational needs of the bakery efficiently and accurately.

No.	Functional Requirements	Description
1	Secure Login and Logout	It allow users to securely access and exit the system using authenticated usernames and passwords.
2	Sales Transaction Processing	This record and process customer purchases efficiently through the Point-of-Sale (POS) module.
3	Product and Price Management	This system shall enable authorized users to manage bakery products, pricing details, and product categories.
4	Real-Time Inventory Monitoring	The system shall continuously monitor inventory quantities and Ingredient stocks to maintain accurate stock records.
5	Supplier and Delivery Tracking	The system shall manage supplier details, purchase orders, and delivery transactions for efficient supply chain operations.
6	Report Generation	The system shall generate accurate sales, inventory, and supply chain reports to support business analysis and decision-making.
7	Low Stock Alerts	The system shall notify users whenever inventory items reach low stock levels to prevent shortages.
8	Customer Transaction Records	The system shall store customer purchase histories and order records for future reference and tracking.

9	User Account and Access Management	The system shall allow administrators to manage user accounts, roles, and system access permissions.
10	Data Backup and Retrieval	The system shall securely back up and retrieve system data to prevent data loss and ensure data recovery when needed.

Functional requirements describe the specific operations, features, and services that the system must perform to meet the operational needs of the bakery. These requirements focus on the core functionalities of Batch Flow, including sales processing, inventory monitoring, supply chain management, and report generation. Functional requirements serve as the foundation of the system because they define how users interact with the application and how the system responds to different business processes and transactions.

In Batch Flow: A POS Based Supply Chain Management System, functional requirements are essential to ensure that bakery operations are organized, efficient, and accurate. The system must support daily activities such as processing customer purchases, updating inventory records, tracking supplier deliveries, and generating business reports in real time. Additionally, the system should provide secure user access, maintain transaction records, and automate monitoring tasks to reduce manual work and human errors. By implementing these functional requirements, the system can improve operational productivity, streamline supply chain activities, and support better decision-making for the bakery business.

Non-Functional Requirements

Non-functional requirements describe the quality standards, performance expectations, and operational characteristics of the system. These requirements ensure that the system is reliable, secure, efficient, and user-friendly for bakery operations and supply chain management.

No.	Non-Functional Requirements	Description
1.	Performance	This process the transactions, inventory updates, and report generation quickly with minimal delay.
2.	Security	This process shall protect user accounts and business data through secure authentication and access control mechanisms.

3.	Reliability	This process shall operate consistently and accurately without frequent system failures or interruptions.
4.	Usability	This process shall provide a user-friendly interface that is easy for administrators, cashiers, and managers to understand and operate.
5.	Scalability	This process shall support future expansion such as additional users, products, suppliers, and branches.
6.	Compatibility	This process shall function properly across different web browsers, devices, and operating systems.

Non-functional requirements define the quality attributes, performance standards, and operational conditions of the system rather than the specific functions it performs. These requirements ensure that Batch Flow operates efficiently, securely, and reliably while supporting the daily business activities of the bakery. They focus on how the system performs in terms of speed, usability, security, reliability, and maintainability to provide a smooth and effective user experience for administrators, cashiers, managers, and suppliers.

In Batch Flow: A POS Based Supply Chain Management System, non-functional requirements are important because the bakery relies on accurate and continuous operations to manage sales, inventory, and supply chain activities. The system must process transactions quickly, protect sensitive business data, and remain accessible during operating hours to avoid delays and operational disruptions. Additionally, the system should be scalable and flexible enough to support future business growth, such as adding more products, users, suppliers, or bakery branches. By meeting these quality standards, the system can improve operational efficiency, ensure data integrity, and provide reliable support for the bakery's daily processes.

CHAPTER 4

SYSTEM DEVELOPMENT AND TESTING

Development Tools and Technologies

The development of Batch Flow: A POS Based Supply Chain Management System utilizes various tools and technologies to ensure efficient system design, development, implementation, and maintenance. These technologies are essential in building a reliable, secure, and user-friendly web-based system capable of handling bakery sales transactions, inventory monitoring, supply chain management, and report generation.

No.	Technology	Purpose
1	HTML	Used for creating the structure of the web pages.
2	CSS	Used for designing and styling the user Interface.
3	JavaScript	Used for interactive features and client-side functionalities.
4	MySQL	Used to store, organize, manage, and retrieve the system data.
5	MongoDB	Used for storing user, product, transaction, and inventory data.

Programming languages, database management systems, web frameworks, and development tools work together to support the functionality and performance of the system. Furthermore, the selected technologies help improve system scalability, data security, processing speed, and overall operational efficiency to meet the needs of the bakery business.

System Implementation

The implementation of Batch Flow: A POS Based Supply Chain Management System followed a structured and systematic development process to ensure that the system effectively met the operational needs of the bakery business. The system was designed to automate sales transactions, inventory monitoring, supplier management, and report generation through a web-based platform. During the planning stage, the researchers identified the major problems encountered in traditional bakery operations, such as manual inventory tracking, delayed sales reporting, and inefficient supply chain coordination. The system requirements, objectives, and functional specifications were carefully analyzed to create a clear foundation for the development process. The development of the system utilized the Agile Software Development Methodology, which focuses on iterative development, continuous improvement, and collaboration among developers and users. The system was divided into different modules, including user authentication, Point-of-Sale (POS), inventory management, supplier tracking, and report generation. Each module was developed and tested incrementally to ensure proper functionality and integration with other components of the system. Regular evaluations and feedback were conducted throughout the development process to identify necessary improvements and ensure that the system aligned with the operational requirements of the bakery.

During the coding and development phase, various programming tools, frameworks, and database technologies were used to build the system. The developers implemented a web-based interface that provides easy navigation and user-friendly interaction for administrators, cashiers, and managers. The database was developed using MySQL to securely store transaction records, inventory data, supplier information, and user accounts. Additionally, security measures such as user authentication and access control were integrated into the system to protect sensitive business information and prevent unauthorized access. Testing and quality assurance were conducted to evaluate the functionality, reliability, and performance of the system before deployment. Different testing methods, including unit testing, integration testing, and user acceptance testing, were performed to identify and resolve errors or system issues. The researchers verified that the system could process sales transactions accurately, update inventory levels in real time, and generate reports efficiently. Furthermore, the system was tested across different devices and web browsers to ensure compatibility, responsiveness, and usability for all intended users.

After successful testing and evaluation, the system was deployed within the bakery environment for actual implementation and operational use. The deployment process included database setup, server configuration, user account creation, and system installation on the required devices. Users such as administrators and cashiers were provided with orientation and guidance on how to operate the system effectively.

Moreover, backup and recovery mechanisms were established to ensure data protection and business continuity in case of technical failures. Through proper implementation and deployment, Batch Flow: A POS Based Supply Chain Management System became capable of improving operational efficiency, minimizing manual work, and supporting better decision-making within the bakery business.

Testing and Evaluation

Test Case

Step #	Action Description	Input Data / Test Parameters	Expected System Behavior & Response	Status
1	Navigate to the main application access portal.	Load `login.html` in browser canvas.	Form accessibility fields load cleanly; input handlers clear previous states.	**PASSED**
2	Input credentials matching the master administrator profile.	Username: `admin` Password: *(Your Password)*	The fixed query bypasses dropdown reliance. Validation maps true profile data attributes.	**PASSED**
3	Initiate authentication execution.	Click the **"Sign In"** button widget.	Server accepts arguments. Browser context updates pathing directly to `management.html`.	**PASSED**
4	Open the Account Registry Control Interface.	Click the **"Manage Users"** toolbar option.	An overlay modal displays. `loadUsers()` launches background requests tracking database operator nodes.	**PASSED**

Step #	**Action Description**	**Input Data / Test Parameters**	**Expected System Behavior & Response**	**Status**
5	Inspect your own active administrator profile row.	Observe row containing the username: `admin (You)`	The system automatically applies a `Protected` design block badge. The red trash icon/button is completely hidden/disabled.	**PASSED**
6	Attempt to trigger a deletion script against another account.	Click **"🗑 Delete"** next to the user row for `baker_john`.	A browser confirmation modal asks: *"Are you absolutely sure you want to permanently delete user account 'baker_john'?"*	**PASSED**
7	Refuse secondary authorization prompt to check stability.	Click **"Cancel"** on the pop-up warning alert.	Deletion loop aborts cleanly. The profile data record remains intact inside the scroll directory view.	**PASSED**
8	Confirm data persistence operations execute reliably.	Click **"🗑 Delete"** on `baker_john` again, then click **"OK"**.	Server reads request metadata context headers, executes safe row deletion, and alerts: *"User profile 'baker_john' successfully removed..."*	**PASSED**

Functionality Testing

Functionality Testing is a software testing process used to verify whether the functions and features of the system operate according to the specified requirements. In Batch Flow: A POS Based Supply Chain Management System, functionality testing ensures that important modules such as login authentication, inventory management, sales processing, supplier management, and report generation work correctly and efficiently. This testing confirms that each function produces the expected output when valid and invalid inputs are provided.

Furthermore, functionality testing helps identify system errors, incorrect computations, and broken features before deployment. Through this testing process, the developers were able to ensure that the system performs accurately and supports the daily operations of the bakery business without interruptions. The successful completion of functionality testing indicates that the system is reliable and capable of meeting user requirements.

User Acceptance Testing

User Acceptance Testing (UAT) is the final phase of software testing where the actual users evaluate the system to determine whether it satisfies their business needs and operational requirements. In Batch Flow, the users such as administrators, cashiers, managers, and suppliers tested the system by performing real-world tasks including sales transactions, inventory updates, and report generation.

The purpose of UAT is to ensure that the system is user-friendly, efficient, and ready for deployment in the bakery environment. Feedback gathered from users helped identify minor improvements and confirmed that the system functions effectively according to expectations. After successful user acceptance testing, the system was considered acceptable for implementation and daily operational use.

Bug Fixes

Bug Fixes refer to the process of identifying, correcting, and improving errors or defects found during system testing and development. During the development of Batch Flow, several issues such as incorrect inventory calculations, delayed report generation, login validation errors, and transaction processing bugs were detected and resolved by the developers.

The bug fixing process improved the system’s overall performance, reliability, and security. Continuous debugging and testing ensured that the identified problems were corrected before the final deployment of the system. As a result, Batch Flow became more stable, accurate, and efficient for managing bakery operations and supply chain activities.

Survey Evaluation Results

The Survey Evaluation Results present the feedback and assessment gathered from respondents who tested and evaluated Batch Flow: A POS Based Supply Chain Management System. The evaluation focused on important criteria such as usability, functionality, reliability, efficiency, security, and user satisfaction. Respondents included bakery staff, managers, and potential users of the system.

Based on the survey results, most respondents rated the system as “Excellent” or “Very Satisfactory” because of its user-friendly interface, accurate inventory tracking, efficient sales monitoring, and reliable report generation. The results indicate that the system effectively improves bakery operations and helps streamline supply chain management processes. Overall, the evaluation confirmed that Batch Flow is acceptable, functional, and beneficial for bakery businesses.

In addition, the survey evaluation revealed that the respondents appreciated the system’s ability to reduce manual work and minimize human errors in bakery operations. Users found the automated inventory monitoring and sales recording features helpful in improving productivity and decision-making. The respondents also highlighted that the system provides faster transaction processing and better organization of supply chain activities, which contributes to smoother daily business operations. These positive responses demonstrate that Batch Flow not only meets the operational requirements of bakery businesses but also enhances overall efficiency, accuracy, and customer service quality.

Moreover, the respondents recommended the implementation of additional features and future enhancements to further improve the system’s capabilities. Suggested improvements included mobile accessibility, automated supplier notifications, barcode scanning, and advanced data analytics for sales forecasting and inventory management. Despite these suggestions, the respondents expressed confidence in the current performance of Batch Flow and agreed that the system provides a practical and effective solution for modern bakery operations. The survey findings strongly support the system’s readiness for deployment and its potential to contribute to the growth and operational success of bakery businesses.

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

Conclusion

In conclusion, the development of Batch Flow: A POS Based Supply Chain Management System successfully achieved its goal of creating an efficient and reliable web-based platform for managing bakery operations. The system integrated essential business processes such as point-of-sale transactions, inventory monitoring, supply chain coordination, sales tracking, and order management into a single centralized platform. Through the implementation of modern web technologies and an Agile development methodology, the project was able to provide a user-friendly and responsive system that improves the overall operational workflow of the bakery business. The system also enhanced data accuracy, reduced manual workloads, and provided real-time access to important business information.

The project addressed several common problems experienced in traditional bakery management processes. Before the implementation of the system, business operations often relied on manual recording methods that were prone to human error, delayed reporting, inventory shortages, and inefficient communication between suppliers, managers, and cashiers. Batch Flow solved these issues by automating sales recording, inventory updates, supply monitoring, and report generation. The system also improved transaction speed and ensured better stock management by providing real-time inventory tracking and automated notifications for low stock levels. As a result, the bakery business can minimize product wastage, avoid overstocking, and maintain a more organized operational process.

Furthermore, Batch Flow provides significant benefits to both management and employees by increasing productivity, improving decision-making, and enhancing customer service. Managers can easily monitor sales performance, inventory status, and supply chain activities through generated reports and dashboards, allowing them to make informed business decisions. Cashiers benefit from a faster and more accurate transaction process, while suppliers can coordinate orders and deliveries more efficiently. Customers also experience improved service due to quicker transactions and better product availability. Overall, the system contributes to a more effective, organized, and sustainable bakery operation, making Batch Flow a valuable solution for modern bakery business management.

Recommendations

For future improvements, it is recommended that Batch Flow: A POS Based Supply Chain Management System continue to enhance its functionality and adaptability to meet the growing demands of bakery businesses. The developers may improve the system's performance by optimizing database management, increasing processing speed, and strengthening system security to protect sensitive business and customer information. Regular system maintenance and updates should also be implemented to ensure compatibility with new technologies and devices. In addition, improving the user interface and user experience can make the system more accessible and easier to navigate for administrators, cashiers, managers, suppliers, and customers.

Additional features may also be integrated to further improve the efficiency and capabilities of the system. One recommended enhancement is the inclusion of an online ordering and reservation feature that allows customers to place orders through the web system remotely. The integration of digital payment gateways such as GCash, Maya, and online banking can also provide more convenient payment options for customers. Furthermore, the system may include SMS or email notification features for order confirmations, low stock alerts, and supplier updates. Advanced analytics and forecasting tools can also be added to help managers predict sales trends, monitor customer preferences, and improve inventory planning.

Possible upgrades for the system include the development of a mobile application version to provide easier access and real-time monitoring through smartphones and tablets. Cloud-based storage and backup solutions may also be implemented to improve data accessibility, scalability, and disaster recovery capabilities. Another valuable upgrade is the integration of barcode scanning and QR code technology to speed up inventory management and cashier transactions. In the future, the system may also incorporate Artificial Intelligence features for automated demand forecasting, smart inventory recommendations, and customer behavior analysis. These enhancements and upgrades can help Batch Flow become a more advanced, efficient, and competitive bakery management solution.

APPENDICES

Appendix A: Survey Form

Name (Optional): ________________________

Age: ________________________

Position/Role: ________________________

Date: ________________________

System Evaluation Questionaire

Criteria	5 Strongly Agree	4 Agree	3 Neutral	2 Disagree	1 Strongly Disagree
The system is easy to use and navigate?					
The system processes transactions quickly?					
The inventory management feature is accurate and reliable?					
The reports generated by the system are useful?					

The system improves bakery operations efficiency?					
The user interface is visually organized and understandable?					
The supply chain management feature is effective?					
The system helps reduce manual workload?					
The system provides accurate sales monitoring?					
Overall, Are you satisfied with the system?					

Suggestions/Comments:

Appendix B: Interview Question

1.What are the common challenges you experience in managing daily bakery operations?

2.How do you currently monitor your inventory and product supplies?

3.What problems do you encounter during sales transactions and customer service?

4.How do you manage orders from customers and suppliers?

5.What features would you like to have in a bakery management system?

6.How important is real-time sales and inventory monitoring in your business operations?

7.How do you handle product shortages, overstocking, or expired ingredients?

8.What methods do you currently use for recording sales and generating reports?

9.Do you think a POS-based supply chain management system can improve your bakery operations? Why or why not?

10.What improvements or technological solutions would help your bakery become more efficient and productive?

Appendix C: User Manual

The Batch Flow user manual provides step-by-step guidance for each user role in the system.

For Admin: (1) Manage Users it includes add new users such as cashiers, managers, and suppliers, edit user information and account permissions, and remove inactive accounts. (2) Manage Inventory this includes add new bakery products and ingredients, update stock quantities and product details, and monitor low-stock notifications. (3) Generate Reports it includes to view sales reports and inventory reports, print or download transaction summaries, and monitor daily, weekly, and monthly performance.

For Cashier: (1) Process Sales Transactions this includes select products purchased by the customer, enter the quantity of items, confirm the total amount, receive payment and complete the transaction, and print or send the receipt digitally. (2) Manage Payments it includes accept cash payments.

For Customer: (1) Place Orders includes browse available bakery products, select desired items, add products to the cart, confirm order and payment details, and submit the order request.

For Supplier: (1) View Purchase Orders this includes access pending and approved purchase orders, confirm delivery schedules and product availability. (2) Update Delivery Status it includes mark orders as shipped or delivered, and notify the bakery regarding supply updates.

For Manager: (1) Daily Sales Summary includes view sales reports and Print or Export Reports, (2) Inventory Status includes monitor inventory and receive low stock alerts. (3) Low Stock Alerts. (4) Recent Transactions. (5)Supply Deliveries. (6) Employee Activity Reports.

Appendix D: Documentation Photos

Sophie’s Mom

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